

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	LTVIP2026TMIDS89849
Project Name	dog breed identification using transfer learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	User Login	Login using email and password Forgot password and reset option
FR-4	Image Upload	Upload dog image from device Validate image format and size
	Image Preprocessing	Resize and normalize image Convert image into model input format
	Dog Breed Prediction	Predict dog breed using transfer learning model Generate confidence score

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should be simple, user-friendly, and accessible to all users with minimal technical knowledge.
NFR-2	Security	Secure authentication, password encryption, and secure data storage should be implemented
NFR-3	Reliability	The system should provide accurate and consistent predictions with minimal errors.
NFR-4	Performance	The system should process and predict dog breeds quickly with low response time.
NFR-5	Availability	The system should be available 24/7 with minimal downtime.
NFR-6	Scalability	The system should support future expansion such as mobile applications and cloud deployment.